Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

Python engineer with 7+ years shipping production systems at scale, now deploying LLM, RAG, and agentic AI applications into production workflows. Currently integrating Anthropic Claude into financial data services at Intuit. Built **Leasel** -- an 8-agent LangGraph compliance platform with a CI eval harness (93.6% precision, 87.9% F1, 1,240 test cases) -- and **AgentProbe**, an open-source LLM evaluation toolkit routing across 5 providers with LLM-as-judge CI. Strong in Python and TypeScript; hands-on with fine-tuning (HuggingFace, PyTorch), RAG pipelines, and multi-agent orchestration. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026). Case studies: AgentProbe · Leasel

CORE COMPETENCIES

LLM Deployment | RAG Pipelines | LLM Evaluation / Evals | Multi-Agent (LangGraph) | Fine-tuning (HuggingFace, PyTorch) | Python (9+ yrs) | TypeScript / Node.js | FastAPI | AWS / Azure / GCP | Docker / Kubernetes | NER / Information Extraction | Production Debugging

EXPERIENCE

Intuit

Jun 2026 - Present

Software Engineer · Virginia, USA

- Deploy Anthropic Claude into production financial data workflows -- integrate LLM APIs, design structured-output schemas, and build eval harnesses (LLM-as-judge + golden datasets) to block hallucinations before release.
- Ground Claude responses in customer financial data using retrieval pipelines; every answer traceable to its source record.
- Build ML pipelines with scikit-learn and XGBoost: data ingestion, feature engineering, model evaluation, and prediction APIs.
- Write guardrails around LLM calls -- input validation, sensitive-data handling, output checks -- and containerize services with Docker for GitHub Actions CI/CD delivery to AWS with production monitoring.
- Develop FastAPI backends and React/TypeScript front ends; ship eval regression tests to catch model quality drops before production release.

Entrata

Oct 2020 - Jul 2024

Senior Software Engineer · Multi-tenant SaaS, 100K+ monthly active users

- Owned 20+ features end-to-end (Python, PHP, Java, JavaScript, React, PostgreSQL) across 120+ Agile sprints.
- Maintained the annual HUD income-limit import for 7 years -- transforming federal source data into ~180,000 domain records powering eligibility, rent, and compliance workflows for 100K+ housing units; zero compliance violations.
- Built a RabbitMQ async processing system for rent updates across thousands of tenant records; retry logic and idempotency eliminated double-processing in production.
- Ran Python/Bash automation cutting manual ops work by 80%; built internal debugging tools reducing triage time by 40%.
- Optimized 200+ SQL queries with execution plan analysis and indexing -- 30% average performance gain across a multi-tenant production database.
- Led Docker microservices redesign (50% better scalability, 30% faster deployments); owned AWS S3 PHI migration with Libsodium encryption -- 100% compliance, zero incidents.

Entrata India

Sep 2018 - Oct 2020

Junior Software Engineer · Pune, India

- Built affordable housing and property management modules (Python, PHP, Java, PostgreSQL); delivered large-scale data import and reconciliation workflows (1.6M HUD/Tax Credit records).
- Led AWS S3 cloud migration from legacy server storage; created 56 state-specific housing certification documents.

PROJECTS

Leasel -- Multi-Agent AI Platform with RAG github.com/venkatalakshmiparimala/Lease_complaince_rag

93.6% precision, 87.9% F1 on 1,240 eval cases. Built an 8-agent LangGraph StateGraph (triage, retrieval, reasoning, validation + HITL escalation) for lease compliance -- proves it beats baseline before deployment. RAG pipeline: OCR ingestion, section-aware chunking, hybrid TF-IDF + semantic reranking, answers cited to exact source clause. Guardrails in code route every eligibility and financial decision to staff; LLM reasons, deterministic logic decides. LLM-as-judge CI harness runs on every PR.

Python · LangGraph (8-agent) · FastAPI · Hybrid RAG (TF-IDF + semantic) · OCR · SQLite · CI/CD · LLM-as-Judge

AgentProbe -- Open-Source LLM Evaluation Toolkit github.com/venkatalakshmiparimala/AgentProbe

15-endpoint FastAPI backend routing prompts across 5 LLM providers (Anthropic, OpenAI, Azure OpenAI, Groq, Ollama). Scores outputs on factual accuracy, hallucination rate, retrieval quality, completeness, and compliance using LLM-as-judge. CI regression tests with golden datasets; used to evaluate Leasel across 1,240+ test cases. Runs 100% locally at zero cost.

Python · FastAPI · LangGraph · ChromaDB · GitHub Actions · Anthropic · OpenAI · Azure OpenAI · Groq · Ollama

AI on FHIR -- Clinical NLP & LLM API Integration github.com/venkatalakshmiparimala/AI_on_FHIR

Turns English/Spanish questions into FHIR API calls using LLMs, BERT, and embeddings -- cut manual clinical lookup time by ~60%. Role-based access control for HIPAA-style data isolation; containerized with Docker.

Python · Flask · React · TypeScript · Docker · BERT · spaCy · LLMs · FHIR REST APIs · NER

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia May 2026

GPA 3.87 · Coursework: custom classification model with HuggingFace + PyTorch (fine-tuning), Isolation Forest anomaly detection, regression modeling (R^2=0.977), NLP pipelines

B.Tech Computer Science & Engineering — JNTU Anantapur 2017

TECHNICAL SKILLS

AI & LLMs: LLM APIs (Anthropic Claude, OpenAI, Azure OpenAI, Groq, Ollama), RAG Pipelines, Hybrid Search, Embeddings, Prompt Engineering, AI Agents, LangGraph, LangChain, Human-in-the-Loop, LLM-as-Judge Evaluation, Fine-tuning (HuggingFace, PyTorch), MCP

Vector Search & NLP: ChromaDB, Pinecone, Weaviate, Elasticsearch, NER, Information Extraction, Semantic Search, BERT, spaCy, HuggingFace

Python & ML: Python (9+ yrs), FastAPI, Flask, Django, Pandas, NumPy, scikit-learn, XGBoost, PyTorch, PySpark

Other Languages & Web: Java, Spring Boot, TypeScript, JavaScript, Node.js, React, Redux, PHP, GraphQL, SQL, PL/SQL, Bash

Databases & Data: PostgreSQL, MySQL, Oracle, Redis, MongoDB, SQLite, Snowflake, ETL Pipelines, Data Modeling, Query Optimization

Cloud & DevOps: AWS (S3, EC2, RDS, Lambda, Glue, SQS, API Gateway, IAM), Azure (OpenAI, Data Factory, Databricks, AKS), GCP (Vertex AI), Docker, Kubernetes, Terraform, GitHub Actions CI/CD

Messaging & Monitoring: RabbitMQ, Kafka, Celery, Prometheus, Grafana, OpenTelemetry

Practices: REST APIs, Microservices, OAuth, Agile/Scrum, TDD, Code Review, HIPAA/PHI Compliance, Observability